Programme : **Diploma in Cloud Computing and Big Data**	
Course Code : **6271A**	Course Title: **Block Chain**
Semester : 6	Credits: **4**
Course Category: **Programme Elective Course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- To understand the history, types and applications of Blockchain
- To understand the essential concepts of blockchain technology
- To describe with Tiers of Blockchain Technology
- To familiarize with Ethereum

Course Prerequisites:

Topic	Course Code	Course name	Semester
Knowledge in programming	3279	Programming In Python	3
Knowledge in DBMS	3272	Database System Concept	3
Knowledge in Web Application Development	4277	Web Application Development lab	3

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the basic notion of distributed systems	14	Understand
CO2	Outline basic concepts of blockchain technology.	15	Understand
CO3	Describe Tiers of Blockchain Technology	15	Understand

CO4	Demonstrate Ethereum Environment	14	Apply
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			2			
CO2	3						
CO3	3						
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Explain the basic notion of distributed systems**		
M1.01	Describe the limitations of Distributed DBMS	2	Understanding
M1.02	Explain the Public Key Cryptography	3	Understanding
M1.03	Outline Hashing, Distributed Systems	3	Understanding
M1.04	Describe the Distributed Consensus & Mining Mechanism	3	Understanding
M1.05	Discuss Generic elements of Blockchain, Features of Blockchain	3	Understanding

Contents:

- Basics: The Double-Spend Problem
- Byzantine Generals' Computing Problems
- Cryptography in Blockchain: Introduction – cryptographic primitives – Asymmetric cryptography, Public Key Cryptography
- Hashing, Distributed Systems
- Distributed Consensus. Mining Mechanism
- Generic elements of Blockchain
- Benefits and limitations ,Features and Types of Blockchain
- Decentralization using blockchain,Methods of decentralization

CO2	**Outline the basic concepts of blockchain technology.**		
M2.01	Explain the Technology Stack	3	Understand
M2.02	Describe Bitcoin Blockchain	3	Understand
M2.03	Summarize Operations and Features of Bitcoin	3	Understand
M2.04	Illustrate Wallets and its Types	3	Understand
M2.05	Explain Consensus Model and Incentive Model.	3	Understand
	Series Test – I	1	

Contents:

- Technology Stack: Blockchain, Protocol, Currency.
- Bitcoin Blockchain:Transactions and Types,Bitcoin White Paper
- Structure of Block, Structure of Block Header
- Operations, Features of Blockchain
- Wallets and Types
- Consensus Model and Incentive Model.

CO3	**Describe Tiers of Blockchain Technology**		
M3.01	Illustrate Blockchain Architecture	3	Understand
M3.02	Explain Blockchain 1.0, Blockchain 2.0, Blockchain 3.0	3	Understand
M3.03	Describe Blockchain Network and Nodes	3	Understand
M3.04	Compare Public Blockchain and Private Blockchain	3	Understand

M3.05	Outline Semi-Private Blockchain, Side chains	3	Understand

Contents:

- Blockchain Architecture – Block, Hash, Distributer P2P
- Tiers of Blockchain Technology: Blockchain 1.0
- Blockchain 2.0, Blockchain 3.0
- Blockchain Network,Nodes-Types of Nodes, Peer-to-Peer Network-Architecture,Uses
- Types of Blockchain: Public Blockchain, Private Blockchain, Semi-Private Blockchain, Side chains.Use Case of BlockChain(In Banking & Medical Records)

CO4	Demonstrate Ethereum Environment		
M4.01	Use web 3.0 based development	3	Apply
M4.02	Explain Ethereum Blockchain, Structure, Operations	2	Understand
M4.03	Describe Smart Contracts	2	Understand
M4.04	Practise Consensus Model, Proof of Stake, Proof of Work	4	Apply
M4.05	Outline Byzantine Fault Tolerance	4	Understand
	Series Test – II	1	

Contents:

- Web 3.0 based development concepts- overview
- Ethereum Blockchain, Elements of the Ethereum block chain
- Ethereum Structure, Operations, Networks
- Smart Contracts
- Proof of Stake, Proof of Work
- Byzantine Fault Tolerance

Text / Reference

T/R	Book Title/Author
T1	Antony Lewis, “The Basics of Bitcoins and Blockchains”
T2	Don Tapscott, “Blockchain Revolution”
T3	Kirankalyan Kulkarni, “Essentials of Bitcoin and Blockchain”, Packt Publishing
T4	Anshul Kaushik, “Block Chain & Crypto Currencies”, Khanna Publishing House.
R1	Andreas M. Antonopoulos , “Mastering Bitcoin: Unlocking Digital Cryptocurrencies”, O’Reilly Media Inc, 2015

R2	Imran Bashir, “Mastering Blockchain: Deeper insights into decentralization, cryptography, Bitcoin, and popular Blockchain frameworks”, Packt Publishing (2017).

Online Resources

Sl.No	Website Link
1	https://kbaiiitmk.medium.com
2	https://learn.kba.ai/
3	https://www.velmie.com/practical-blockchain-study
4	https://blockchainhub.net/blockchains-and-distributed-ledger-technologies-in-general/